AthenaStudies

Linear Algebra and Calculus

Calculus Midterm – Functions and Limits

Structure

- Structure based on past exams
- Learn the most by asking questions
- Active learning with old exam questions is the basis of our trainings
- 24/7 WhatsApp aid

Agenda

1. **Functions**
2. **Limits**
3. Continuity
4. Derivatives

Part 1: Functions

Functions

Definition (P.4.1). Let D and S be sets. A **function** $f : D \to S$ is a rule that assigns a *unique* element $f(x)$ in S to each element in D.

Usually also written as

$$\begin{aligned} f : & D \to S \\ & x \to f(x) \ . \end{aligned}$$

- A **function** *f* assigns each input exactly one output
  - D is the domain of *f(D(f))*
    - Domain D: all inputs for which the function is defined
    - "Allowed inputs"
    - Domain is defined by definition
  - If $x \in D$, then f(x) is the image of x
    - Image f(x) is the actual output for the input x
  - The set of all images $x \in D$ is called the range of f(R(f))
    - All the actual outputs
    - The range is defined by computation

Unless specified, functions will be $f : R \to R$

Functions

Example. Some examples of real functions are

- $f(x) = x^2$, with $f(2) = 4, f(-2) = 4, f(3) = 9$. Its domain is $\mathbb{R}$ and the range is $\mathbb{R}_{\geq 0}$.
- $f(x) = |x|$, with $f(-3) = 3, f(2) = 2, f(0) = 0$. Its domain is $\mathbb{R}$ and the range is $\mathbb{R}_{\geq 0}$.
- $f(x) = \frac{1}{x}$, with $f(1) = 1, f(2) = \frac{1}{2}, f(-5) = -\frac{1}{5}$. Its domain is [redacted] and the range is [redacted].

Example 1: $f(x) = x^2$

- Domain D(f) = all real numbers, because you can square any real number
  - Remember, D is all the allowed inputs
- Range R(f) = all real numbers that are greater than or equal to zero
  - Squaring a number never produces a negative output

Example 2: $f(x) = |x|$

- $|x|$ gives the absolute value of x (the distance of x from 0 on the number line)
- Domain D(f) = all real numbers, because you can take the absolute value of any real number
- Range R(f) = all real numbers that are greater than or equal to zero
  - Absolute value can never be negative

Functions

Example. Some examples of real functions are

- $f(x) = x^2$, with $f(2) = 4, f(-2) = 4, f(3) = 9$. Its domain is $\mathbb{R}$ and the range is $\mathbb{R}_{\geq 0}$.
- $f(x) = |x|$, with $f(-3) = 3, f(2) = 2, f(0) = 0$. Its domain is $\mathbb{R}$ and the range is $\mathbb{R}_{\geq 0}$.
- $f(x) = \frac{1}{x}$, with $f(1) = 1, f(2) = \frac{1}{2}, f(-5) = -\frac{1}{5}$. Its domain is $\mathbb{R}\backslash\{0\}$ and the range is $\mathbb{R}\backslash\{0\}$.

Example 3: f(x) = 1/x

- The function returns the reciprocal of a number

- Domain D(f) = all real numbers \ {0}
  - \ {0} means “excluding” 0
  - Why? 0 is an invalid input as x is in the denominator, and we can’t divide by zero

- Range R(f) = all real numbers \ {0}
  - Because there is no real number x such that 1/x = 0

Operations on Functions

Definition (P.5.3). If f, g are functions and x is in both their domains, then

- $(f+g)(x) = f(x) + g(x)$
- $(f-g)(x) = f(x) - g(x)$
- $(fg)(x) = f(x)g(x)$
- $(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$, when $g(x) \neq 0$.

Remember, we cannot divide by zero, and g(x) is the domain of f

Composition of Functions

Definition (P.5.4). If $f : D \to S$ and $g : R \to D$ are functions, then $f \circ g : R \to S$ is the **composition** such that $(f \circ g)(x) = f(g(x))$. The domain of $f \circ g$ is made up of the values in the domain of g whose image is in the domain of f.

Remember:

- Domain of f is D (allowed inputs), codomain of f is S (potential outputs)
- Domain of g is R (allowed inputs), codomain of g is D (potential outputs)

Here, we see that the outputs of g lie inside D, which is the domain of f.

We can rewrite this as:

$$R \xrightarrow{g} D \xrightarrow{f} S$$

= R → S

Which gives us the composition (f o g)(x) = f(g(x))

- Which has the type f o g: R → S

Interpretation: Apply g first, then apply f to the result of g. Composition here only works because the codomain of g is equal to the domain of f.

If we have a composition f(g(x)), where the output of g(x) cannot be an input to f, the function will be invalid.

Composition of Functions

Example. Let $f(x) = \cos x$ and $g(x) = x^3$. Then $(f \circ g)(x) = \cos(x^3)$.

- Function f takes a real number and returns its cosine
  - f: R → R
  - f(x) = cos x
- Function g takes a real number, and cubes it
  - G: R → R
  - G(x) = x^3

Here, the domains work as g(x) outputs real numbers, and f(x) accepts any real number as input. The codomain of g = domain of f.

When we have f(g(x)), this means we apply g(x) first, then apply f to the result. $x \xrightarrow{g} x^3 \xrightarrow{f} \cos(x^3)$

We get (f o g)(x) = cos(x^3)

Piecewise Defined Functions

Piecewise functions take on different formulas on different parts of the domain.

Example.

$$f(x) = \begin{cases} x^2, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Here, if the input x is greater than or equal to zero, we apply the formula x^2 to it.

However, if the input x is less than zero, we apply the formula –x to it.

Polynomial Functions

Definition (P.6.5). A **polynomial** is a function $P : \mathbb{R} \to \mathbb{R}$ such that $P(x) = a_0 + a_1 x + \cdots + a_n x^n$, where $a_0, a_1, \cdots, a_n \in \mathbb{R}$ are the *coefficients* with $a_n \neq 0$ if $n > 0$. The number n is called the **degree** of the polynomial.

- A polynomial is a function made by adding together powers of x multiplied by numbers (coefficients).
- a_n does not equal to zero **if** n > 0
- n is called the *degree of the polynomial*
  - The degree is the highest power with a non-zero coefficient.

Polynomial Functions

Example. Some examples are

- $P(x) = 3$ is a polynomial of degree 0.
- $P(x) = 2 - x$ is a polynomial of degree 1.
- $P(x) = 3x^3 + 2x^2 - 1$ is a polynomial of degree [redacted]

Example 1:

- P(x) = 3 is a constant with no visible x. However, we can rewrite it as:
  - $3 = 3x^0$
  - x^0 will always equal 1 no matter the value of x. 3*1 = 3
  - So, n here is equal to 0, meaning P(x) = 3 has degree 0

Example 2:

- P(x) = 2-x has a visible x, which can be rewritten as x^1
  - This means n is equal to 1, meaning P(x) = 2-x has degree 1

Polynomial Functions

Example. Some examples are

- $P(x) = 3$ is a polynomial of degree 0.
- $P(x) = 2 - x$ is a polynomial of degree 1.
- $P(x) = 3x^3 + 2x^2 - 1$ is a polynomial of degree 3.

Example 3:

- $P(x) = 3x^3 + 2x^2 - 1$
  - The highest power of x in this function is 3, meaning the polynomial is of degree 3

Polynomial Functions

Proposition. *Let P be a polynomial of degree n and Q a polynomial of degree m with $n \neq m$. Then:*

- *$P + Q$ has degree $\max\{n, m\}$.*
- *PQ has degree $n + m$.*

If we are adding two polynomial functions P with degree n and Q with degree m, where n does not equal to m;

- We take the maximum of {n, m} (the bigger degree) as that will be the degree of function P + Q
- Because $x^3 + x^2 = x^3 + x^2$, meaning the highest power is 3.

However, if we are multiplying the two polynomial functions

- We take n + m as the degree of the function PQ
- Because $x^3 * x^2 = x^5$

Polynomial Functions - Important Definitions

Definition (P.6.6). Let P and Q be polynomials with $Q \neq 0$, then $R(x) = \frac{P(x)}{Q(x)}$ is a **rational function** whose domain is $\mathbb{R}$ except for the values x such that $Q(x) = 0$.

We cannot divide by 0

Example. The domain of $R(x) = \frac{3x^2+4}{2x-1}$ is $\mathbb{R}\backslash\{\frac{1}{2}\}$ since $2x - 1 = 0$ if and only if $x = 1/2$.

Definition (P.6). A zero of a function f is a value $r \in \mathbb{R}$ such that $f(r) = 0$.

A zero of a polynomial is also called *root*.

Example. A zero of $f(x) = x^3 - 3x^2 + 2$ is $x = 1$ since $f(1) = 1^3 - 3 \cdot 1^2 + 2 = 0$.

Equate the function to zero and solve to get the value of x
for which the function returns zero

Polynomial Functions

Theorem (P.6.1). ***The Factor Theorem*** *The number r is a root of the polynomial P of degree not less than 1 if and only if $x - r$ is a factor of $P(x)$.*

Remember, the root of a polynomial is a number r where P(r) = 0.

The factor theorem states that if r is a root of the polynomial function P(x), then the polynomial must contain the factor (x–r). And if (x-r) is a factor, plugging in r will give 0.

Polynomial Functions

Theorem (P.6.1). ***The Factor Theorem*** *The number r is a root of the polynomial P of degree not less than 1 if and only if $x - r$ is a factor of $P(x)$.*

Example. From the previous example, $f(x) = x^3 - 3x^2 + 2 = (x-1)(x^2 - 2x - 2)$.

Sometimes, it is useful to factor a polynomial, for instance, when you know a root and want to know the rest. This can come in handy in certain situations, for example, for computing eigenvalues of matrices (more on that later on in the course). Here are some simple factoring rules:

- $ax^2 + bx = x(ax + b)$.
- $x^2 - a^2 = (x - a)(x + a)$.
- $x^2 + 2ax + a^2 = (x + a)^2$.
- $x^2 - 2ax + a^2 = (x - a)^2$.
- $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$.

The example $f(x) = x^3 - 3x^2 + 2$

Can be rewritten as $(x - 1)(x^2 - 2x - 2)$ using the factoring rules. This gives us the factor (x-1) multiplied by some polynomial = f(x), which means 1 is a root of the polynomial.

If we plug in 1:

$f(x) = 1 - 3 + 2 = 0$

Trigonometric Functions

We measure angles in radians, as the length of the arc that starts at point (1,0) and follows the unit circle counterclockwise (in such units, π radians = 180◦)

Definition (P.7.8)**.** For any real t, the **cosine** of t (abbreviated $\cos t$) and the **sine** of t (abbreviated $\sin t$) are the x- and y-coordinates of the point P_t on the unit circle (where t is the angle formed at the origin by the positive x semiaxis and the segment connecting the origin and the point P_t).

$$\cos t = \text{the } x\text{-coordinate of } P_t$$
$$\sin t = \text{the } y\text{-coordinate of } P_t.$$

We define sin t and cos t using the **unit circle** (a circle of radius 1 centered at the origin).

For any real angle t (in radians), move t units around the unit circle starting at (1,0):

- cos t = the x-coordinate of the point
- sin t = the y-coordinate of the point

Trigonometric Functions

Remark. Notice that we are computing the sine and cosine of an angle in the circle, up to 2π, however, taking a larger angle simply means doing one full turn to the circle and some more. That remaining angle is computed by getting the remainder of the division of the original angle by 2π. This means that trigonometric functions are *periodic* (and hence the wavy shape of their graphs), that is,

$$\cos(t + 2\pi) = \cos t, \quad \sin(t + 2\pi) = \sin t.$$

In cos(t + 2π) and sin(t + 2π):

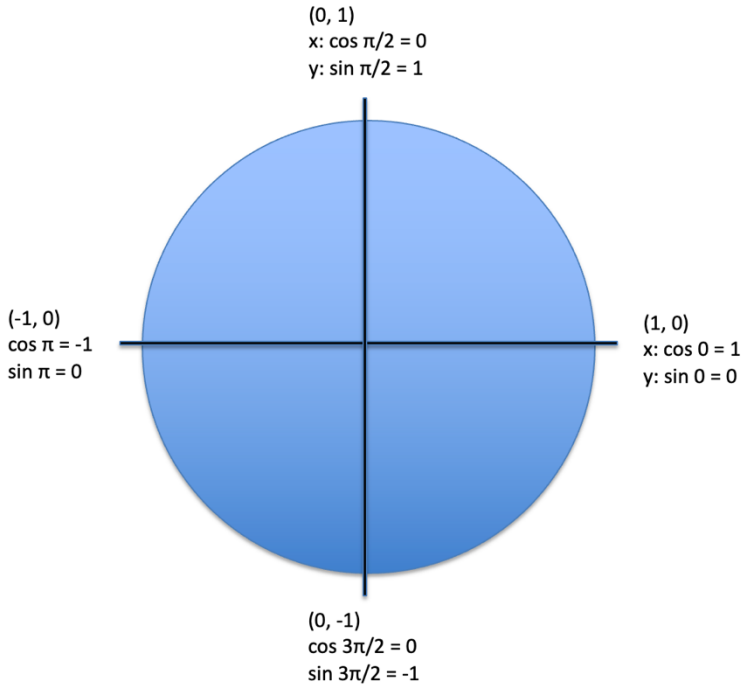
- 2π means one full turn to the circle. So, we can calculate the x and y-coordinates by taking simply cos t and sin t .

Trigonometric Functions

Example.

$$\cos 0 = 1, \quad \cos\frac{\pi}{2} = 0, \quad \cos\pi = -1, \quad \cos\left(-\frac{\pi}{2}\right) = \cos\frac{3\pi}{2} = 0$$

$$\sin 0 = 0, \quad \sin\frac{\pi}{2} = 1, \quad \sin\pi = 0, \quad \sin\left(-\frac{\pi}{2}\right) = \sin\frac{3\pi}{2} = -1.$$



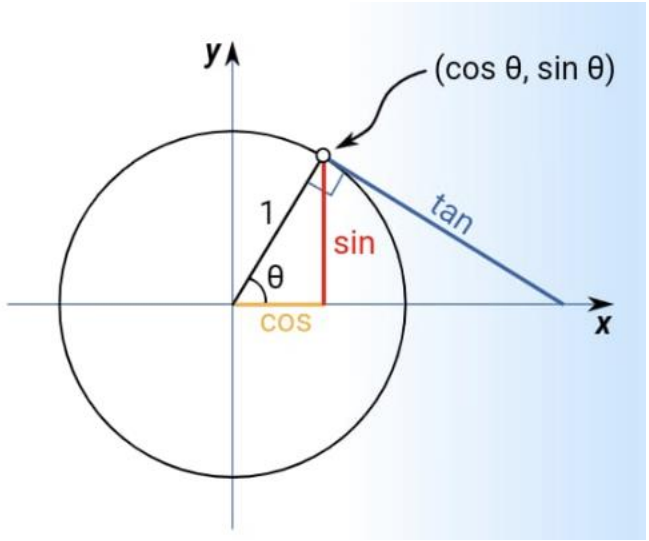
Trigonometric Functions

Definition (P.7.9)**.** The **tangent** of an angle t is defined in terms of sine and cosine as

$$\tan t = \frac{\sin t}{\cos t}.$$

There are some useful identities that may come in handy in some computations:

- The range of both $f(x) = \cos x$ and $g(x) = \sin x$ is $[-1, 1]$, that is, $-1 \leq \cos x, \sin x \leq 1$, for all $x \in \mathbb{R}$.
- $\cos^2 t + \sin^2 t = 1$.
- Cosine is an even function; $\cos(-t) = \cos t$.
- Sine is an odd function; $\sin(-t) = -\sin t$.



Today's Agenda

1. ~~Functions~~
2. **Limits**
3. Continuity
4. Derivatives

Part 2: Limits

Limits

Definition (1.2.1)**.** We say that a function f has limit L at a point $x = a$ if $f(x)$ gets as close as we want to L when we take x close enough to a and write it as

$$\lim_{x \to a} f(x) = L.$$

What value does f(x) get closer to as x gets closer to a?

Example. Limits to some functions:

- $\lim_{x \to a} x^2 = a^2$
- $\lim_{x \to a} c = c$

- The limit L of a function f describes what value the function approaches as the input gets close to a certain point.
- Limits do not care what happens **at** x = a, only what happens **near** x = a

Intuition:

- Limits are about approach, not plugging in
  - Think of a limit as a door
  - It does not matter if the door is open, closed, etc..
  - What matters is that you're heading towards the door
  - So:
    - A function can have a limit even if it is undefined at that point
    - A function can have a limit even if the value at that point is different

Limits

Definition (1.2.1)**.** We say that a function f has limit L at a point $x = a$ if $f(x)$ gets as close as we want to L when we take x close enough to a and write it as

$$\lim_{x \to a} f(x) = L.$$

What value does f(x) get closer to as x gets closer to a?

Example. Limits to some functions:

- $\lim_{x \to a} x^2 = a^2$
- $\lim_{x \to a} c = c$

Examples:

- $\lim_{x \to a} x^2 = a^2$
  - Take f(x) = x^2 , and $\lim_{x \to a} f(x)$. Just plug in $\lim_{x \to a} x^2 = a^2$
- $\lim_{x \to a} c = c$
  - Take f(x) = c , and $\lim_{x \to a} f(x)$. Just plug in $\lim_{x \to a} c = c$

Simpler Example:

- $f(x) = 2x + 1, \quad \lim_{x \to 3} f(x)$
  - Just plug in $\lim_{x \to 3} (2x + 1) = 7$

Limits

Exercise. Find the following limits:

- $\lim_{x \to -2} \frac{x^2 + x - 2}{x^2 + 5x + 6}$
- $\lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$
- $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x^2 - 16}$

Try and solve the above exercise without looking at the answers. Go back to the previous slide if needed.

Limits

Exercise. Find the following limits:

- $\lim_{x \to -2} \frac{x^2 + x - 2}{x^2 + 5x + 6}$
- $\lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$
- $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x^2 - 16}$

Sometimes, it is useful to factor a polynomial, for instance, when you know a root and want to know the rest. This can come in handy in certain situations, for example, for computing eigenvalues of matrices (more on that later on in the course). Here are some simple factoring rules:

- $ax^2 + bx = x(ax + b)$.
- $x^2 - a^2 = (x - a)(x + a)$.
- $x^2 + 2ax + a^2 = (x + a)^2$.
- $x^2 - 2ax + a^2 = (x - a)^2$.
- $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$.

Exercise 1:

- Try direct substitution, plug in x = -2
  - Numerator: $(-2)^2 + (-2) - 2 = 4 - 2 - 2 = 0$
  - Denominator: $(-2)^2 + 5(-2) + 6 = 4 - 10 + 6 = 0$
  - We end up with 0/0, which means we need to factor:
- Factorize
  - Numerator: $x^2 + x - 2 = (x + 2)(x - 1)$
  - Denominator $x^2 + 5x + 6 = (x + 2)(x + 3)$
  - Cancel common factor (x + 2) : we end up with (x-1)/(x+3)
    - (x != -2)
  - Now substitute:

$$\lim_{x \to -2} \frac{x - 1}{x + 3} = \frac{-2 - 1}{-2 + 3} = \frac{-3}{1} = -3$$

  - **Answer = -3**

Limits

Exercise. Find the following limits:

- $\lim_{x \to -2} \frac{x^2 + x - 2}{x^2 + 5x + 6}$
- $\lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$
- $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x^2 - 16}$

Sometimes, it is useful to factor a polynomial, for instance, when you know a root and want to know the rest. This can come in handy in certain situations, for example, for computing eigenvalues of matrices (more on that later on in the course). Here are some simple factoring rules:

- $ax^2 + bx = x(ax + b)$.
- $x^2 - a^2 = (x - a)(x + a)$.
- $x^2 + 2ax + a^2 = (x + a)^2$.
- $x^2 - 2ax + a^2 = (x - a)^2$.
- $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$.

Exercise 2:

- We can combine the numerator: $\frac{1}{x} - \frac{1}{a} = \frac{a - x}{ax}$
  - The whole expression becomes:

$$\frac{\frac{a-x}{ax}}{x - a}$$

  - Factor out the negative (to be able to cancel out of denominator: a - x = -(x – a)
  - So, we end up with

$$\frac{\frac{-(x-a)}{ax}}{x - a} = \frac{-(x - a)}{ax(x - a)} = -\frac{1}{ax}$$

Now plug in the limit x → a

$$\lim_{x \to a} -\frac{1}{ax} = -\frac{1}{a \cdot a} = -\frac{1}{a^2}$$

We get our **answer:** $-\frac{1}{a^2}$

Limits

Exercise. Find the following limits:

- $\lim_{x \to -2} \frac{x^2 + x - 2}{x^2 + 5x + 6}$
- $\lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$
- $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x^2 - 16}$

Sometimes, it is useful to factor a polynomial, for instance, when you know a root and want to know the rest. This can come in handy in certain situations, for example, for computing eigenvalues of matrices (more on that later on in the course). Here are some simple factoring rules:

- $ax^2 + bx = x(ax + b)$.
- $x^2 - a^2 = (x - a)(x + a)$.
- $x^2 + 2ax + a^2 = (x + a)^2$.
- $x^2 - 2ax + a^2 = (x - a)^2$.
- $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$.

Exercise 3:

- Try direct substitution
  - This gives us:
    - Numerator = 2 – 2 = 0
    - Denominator = 16 – 16 = 0
    - This gives us 0/0, meaning we need to simplify
- Factor the denominator
  - $x^2 - 16 = (x-4)(x+4)$
- Simplify the numerator by multiplying both numerator and denominator by $\sqrt{x} + 2$
  - The numerator becomes

$$(\sqrt{x} - 2)(\sqrt{x} + 2) = x - 4$$

Now we can cancel the common factor from the numerator and the denominator

$$\frac{x - 4}{(x - 4)(x + 4)(\sqrt{x} + 2)} = \frac{1}{(x + 4)(\sqrt{x} + 2)}$$

And plug in x → 4

$$\frac{1}{(4 + 4)(\sqrt{4} + 2)} = \frac{1}{8 \cdot 4} = \frac{1}{32}$$

We get our **answer** = 1/32

Limits

Definition (1.2.2)**.** If f is defined on (b, a) (resp. (a, b)), we define the **left** (resp. **right**) **limit** of f as $\lim_{x \to a^-} f(x)$ (resp. $\lim_{x \to a^+} f(x)$) if x gets close to a from the left side (resp. right).

Example. Consider the function

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Then $\lim_{x \to 0^-} \operatorname{sgn} x = -1$ and $\lim_{x \to 0^+} \operatorname{sgn} x = 1$.

Left and Right Limits:

- A left-hand limit $\lim_{x \to a^-} f(x)$
  - Describes the value that f(x) approaches as x gets close to a from the left side (values smaller than a)
- A right-hand limit $\lim_{x \to a^+} f(x)$
  - Describes the value that f(x) approaches as x gets close to a from the right side (values larger than a)
- Tip to remember (plus sign in limit for values larger than a, and negative sign for values smaller than a)

Limits

Definition (1.2.2)**.** If f is defined on (b, a) (resp. (a, b)), we define the **left** (resp. **right**) **limit** of f as $\lim_{x \to a^-} f(x)$ (resp. $\lim_{x \to a^+} f(x)$) if x gets close to a from the left side (resp. right).

Example. Consider the function

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Then $\lim_{x \to 0^-} \operatorname{sgn} x = -1$ and $\lim_{x \to 0^+} \operatorname{sgn} x = 1$.

Example: The Sign Function

- As $x \to 0^-$ (approaching 0 from negative values) => $\lim_{x \to 0^-} \operatorname{sgn} x = -1$
- As $x \to 0^+$ (approaching 0 from positive values) => $\lim_{x \to 0^+} \operatorname{sgn} x = 1.$

Key idea is that left and right limits can be different.

Limits

Theorem (1.2.1).

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.$$

Relationship between two-sided and one-sided limits:

- Theorem: A two-sided limit holds if and only if both one-sided limits exist and are equal to each other and to L
- To have a full limit at x = a
  - The function must approach the same value from the left and right side
- If the left-hand limit and right-hand limit are:
  - Both defined, and
  - Equal,
  - Then the two-sided limit exists and equals that value
- If not:

$$\lim_{x \to a} f(x) \text{ does not exist.}$$

* Note that a function can have different one-sided limits like in the previously slide, except it will not have a two-sided limit.

Limits

Theorem (1.2.1).

$$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

Exercise. Compute the lateral limits at $x = 2$ for $f(x) = \frac{|x-2|}{x^2+x-6}$. What about $\lim_{x\to 2} f(x)$?

Exercise:

- If we try direct substitution, we get 0/0, meaning we need to factorize:
  - Denominator = $x^2 + x - 6 = (x + 3)(x - 2)$
- For the numerator |x-2|, it behaves differently on each side of 2:
  - If $x > 2$, then $|x - 2| = x - 2$
  - If $x < 2$, then $|x - 2| = 2 - x = -(x - 2)$
- So first solve for $x > 2$ (RHL):

$$f(x) = \frac{x-2}{(x-2)(x+3)} = \frac{1}{x+3}$$

$$\lim_{x\to 2^+} f(x) = \frac{1}{2+3} = \frac{1}{5}$$

Then for $x < 2$ (LHL):

$$f(x) = \frac{2-x}{(x-2)(x+3)} = \frac{-(x-2)}{(x-2)(x+3)} = -\frac{1}{x+3}$$

$$\lim_{x\to 2^-} f(x) = -\frac{1}{2+3} = -\frac{1}{5}$$

- Since the RHL and LHL are not equal, this means the two-sided limit does not exist.

Limits

Theorem (1.2.2). *Let f, g be functions such that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$ and $k \in \mathbb{R}$, then*

- $\lim_{x \to a} (f(x) + g(x)) = L + M$
- $\lim_{x \to a} (f(x) - g(x)) = L - M$
- $\lim_{x \to a} (f(x) \cdot g(x)) = L \cdot M$
- $\lim_{x \to a} kf(x) = kL$
- $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, *if* $M \neq 0$
- $\lim_{x \to a} (f(x))^{\frac{m}{n}} = L^{\frac{m}{n}}$, *provided* $L > 0$ *if* n *is even and* m *is odd and* $L \neq 0$ *if* $m < 0$
- *If* $f(x) \leq g(x)$, *then* $L \leq M$.

Limits

Theorem (1.2.4). ***The Squeeze Theorem*** *Let f, g, h functions such that $f(x) \leq g(x) \leq h(x)$ $\forall x \in (b, c)$ and let $a \in (b, c)$. Assume $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$. Then $\lim_{x \to a} g(x) = L$.*

Example. Since $3 - x^2 \leq u(x) \leq 3 + x^2$ $\forall x \neq 0$, $\lim_{x \to 0} 3 - x^2 = \lim_{x \to 0} 3 + x^2 = 3 \implies \lim_{x \to 0} u(x) = 3$.

If a function is “squeezed” between two functions that both approach the same value, it is forced to approach that value too.

In the example:

- $3 – x^2 <= u(x) <= 3 + x^2$ $\forall$!= 0
- That means:
  - Lower bound: $3– x^2$
  - Upper bound: $3 + x^2$
  - u(x) is in between these two functions (squeezed)
  - If we compute the limits of the outer functions, both get 3
  - Therefore, by the squeeze theorem

$$\lim_{x \to 0} u(x) = 3.$$

Limits at Infinity and Infinite Limits

Definition (1.3.3). Let f be defined over (a, ∞), if $f(x)$ is as close to L as we want by taking x large enough, then we say $\lim_{x\to\infty} f(x) = L$. Analogously, $\lim_{x\to-\infty} f(x) = M$.

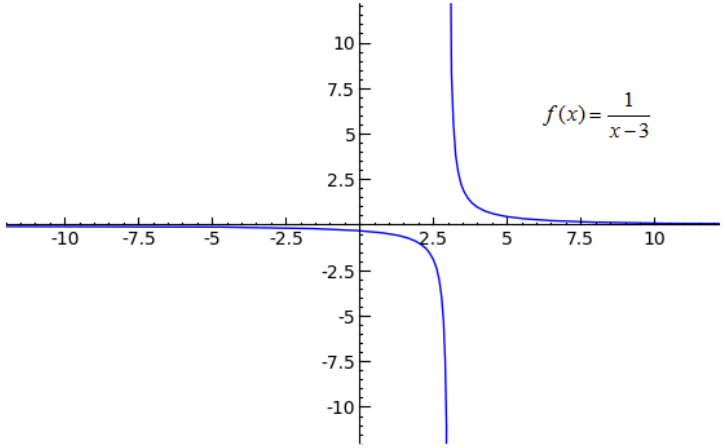
Limit at Infinity:

- If f is defined for all large enough x, and the values of f(x) get arbitrarily close to a number L as x becomes larger, then:

$$\lim_{x\to\infty} f(x) = L.$$

- If f(x) approaches a number M as x becomes very negative, then:

$$\lim_{x\to-\infty} f(x) = M.$$



Think of asymptotes: As x keeps getting larger (closer to infinity) the function approaches L.

* Same goes for negative infinity

Limits at Infinity and Infinite Limits

Definition (1.3.3). Let f be defined over (a, ∞), if $f(x)$ is as close to L as we want by taking x large enough, then we say $\lim_{x\to\infty} f(x) = L$. Analogously, $\lim_{x\to-\infty} f(x) = M$.

Exercise. Calculate $\lim_{x\to\pm\infty} f(x)$, where $f(x) = \frac{x}{\sqrt{x^2+1}}$.

Exercise:

- Square root of x^2 is equal to |x| (the absolute value of x)
- Remember that the absolute value of x gives different behavior for x > 0 and x < 0
- For large |x|, the +1 inside the square root becomes negligible compared to x^2

Solving:

- Factor out x^2 inside the root of the denominator:

$$\sqrt{x^2+1} = \sqrt{x^2\left(1+\frac{1}{x^2}\right)} = |x|\sqrt{1+\frac{1}{x^2}}$$

$$f(x) = \frac{x}{|x|\sqrt{1+\frac{1}{x^2}}}$$

Case 1: x → +infinity

- If x > 0, then |x| = x

$$f(x) = \frac{x}{x\sqrt{1+\frac{1}{x^2}}} = \frac{1}{\sqrt{1+\frac{1}{x^2}}}$$

- As x gets larger, $1/x^2$ approaches zero, so we can replace it with 0 inside the limit

$$\lim_{x\to+\infty} f(x) = \frac{1}{\sqrt{1+0}} = 1.$$

Answer = 1

- Case 2 on next slide

Limits at Infinity and Infinite Limits

Definition (1.3.3). Let f be defined over (a, ∞), if $f(x)$ is as close to L as we want by taking x large enough, then we say $\lim_{x\to\infty} f(x) = L$. Analogously, $\lim_{x\to-\infty} f(x) = M$.

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$$f(x) = \frac{x}{|x|\sqrt{1+\frac{1}{x^2}}}$$

Case 2: x → -infinity

- If x < 0, then |x| = -x

$$f(x) = \frac{x}{-x\sqrt{1+\frac{1}{x^2}}} = -\frac{1}{\sqrt{1+\frac{1}{x^2}}}$$

- Again, as x approaches negative infinity, $1/x^2$ approaches zero, so we can replace it with 0 inside the limit

$$\lim_{x\to-\infty} f(x) = -\frac{1}{\sqrt{1+0}} = -1.$$

Answer = -1

Limits at Infinity and Infinite Limits

Example. For limits toward infinity in rational functions it is important to look for the term of highest degree in the denominator:

- $$\lim_{x\to\pm\infty} \frac{2x^2 - x + 3}{3x^2 + 5} = \lim_{x\to\pm\infty} \frac{\frac{2x^2-x+3}{x^2}}{\frac{3x^2+5}{x^2}} = \lim_{x\to\pm\infty} \frac{2 - \frac{1}{x} + \frac{3}{x^2}}{3 + \frac{5}{x^2}} = \frac{2}{3}$$
- $$\lim_{x\to\pm\infty} \frac{5x + 2}{2x^3 - 1} = \lim_{x\to\pm\infty} \frac{\frac{5}{x^2} + \frac{2}{x^3}}{2 - \frac{1}{x^3}} = 0.$$

For limits as $x \to \pm\infty$, the highest-degree term dominates the behavior of the function.

A good strategy is to divide the numerator and the denominator by the highest power of x in the denominator

General rules / tricks:

If degree(top) < degree(bottom) → limit is **0**

If degree(top) = degree(bottom) → limit is **ratio of leading coefficients**

If degree(top) > degree(bottom) → limit is **±∞** (or does not exist)

Limits at Infinity and Infinite Limits

Example. Limits to points can also lead to infinity:

$$\lim_{x \to 0} \frac{1}{x^2} = \infty.$$

Not all limits approach a finite number, sometimes functions can grow without a bound as x approaches a certain point.

- As x gets closer to 0, the denominator x^2 becomes extremely small
- Dividing by a very small number makes the value extremely large.
- The function increases without bound from both sides of 0.
- This can indicate a vertical asymptote at x = 0

$$\frac{1}{x^2} \to \infty \quad \text{as } x \to 0.$$

Limits at Infinity and Infinite Limits

Example. Careful, recall that if lateral limits don't coincide, the limit does not exist:

$$\lim_{x\to 0^+} \frac{1}{x} = \infty, \quad \lim_{x\to 0^-} \frac{1}{x} = -\infty.$$

Right hand limits:

- If x is positive but very small (e.g., 0.1, 0.01, 0.001),
  - the values 1/x become very large positive numbers.

Left hand limits:

- If x is negative but very small in magnitude (e.g., –0.1, –0.01, –0.001),
  - the values 1/x become very large negative numbers.

* Here the two one sided limits are not equal, remember that this means the two-sided limit does not exist.

Limits at Infinity and Infinite Limits

Example. There are also limits to infinity that lead to infinity

$$\lim_{x\to\pm\infty} \frac{x^4+3x}{x^2-x+5} = \lim_{x\to\pm\infty} \frac{x^2+\frac{3}{x}}{1-\frac{1}{x}+\frac{5}{x^2}} = \infty.$$

When taking limits at infinity of rational functions, the behavior is dominated by the highest-degree terms.

Here:

- Numerator degree = 4
- Denominator degree = 2

Since the numerator grows much faster than the denominator, the fraction will grow without bound → infinity.

If we divide the numerator by the highest power in the denominator:

$$\frac{x^4+3x}{x^2-x+5} = \frac{x^2+\frac{3}{x}}{1-\frac{1}{x}+\frac{5}{x^2}}.$$

$$\frac{3}{x} \to 0$$
$$\frac{1}{x} \to 0$$
$$\frac{5}{x^2} \to 0$$

Limits at Infinity and Infinite Limits

Example. There are also limits to infinity that lead to infinity

$$\lim_{x\to\pm\infty} \frac{x^4+3x}{x^2-x+5} = \lim_{x\to\pm\infty} \frac{x^2+\frac{3}{x}}{1-\frac{1}{x}+\frac{5}{x^2}} = \infty.$$

Proof:

If we divide the numerator by the highest power in the denominator:

$$\frac{x^4+3x}{x^2-x+5} = \frac{x^2+\frac{3}{x}}{1-\frac{1}{x}+\frac{5}{x^2}}.$$

And as x approach +- infinity:

$$\frac{3}{x} \to 0$$
$$\frac{1}{x} \to 0$$
$$\frac{5}{x^2} \to 0$$

So, the expression becomes:

$$\frac{x^2+0}{1-0+0} = x^2$$

and as x approaches +- infinity

$$x^2 \to +\infty.$$

Limits at Infinity and Infinite Limits

Example. We have to be mindful of the signs on certain limits

$$\lim_{x\to-\infty} \frac{x^3-5x+7}{-x^2+3x-2} = \lim_{x\to-\infty} \frac{x-\frac{5}{x}+\frac{7}{x^2}}{-1+\frac{3}{x}-\frac{2}{x^2}} = \infty.$$

If we divide by the highest power in the denominator x^2:

$$\frac{x^3-5x+7}{-x^2+3x-2} = \frac{x-\frac{5}{x}+\frac{7}{x^2}}{-1+\frac{3}{x}-\frac{2}{x^2}}.$$

Take the limit as x → -inf

$$\frac{5}{x} \to 0,\ \frac{7}{x^2} \to 0,\ \frac{3}{x} \to 0,\ \frac{2}{x^2} \to 0$$

So, the fraction becomes:

$$\frac{x}{-1} = -x$$

Watch the sign, as x → -inf, we have –x → +inf, so the whole function grows without bound

$$\lim_{x\to-\infty} \frac{x^3-5x+7}{-x^2+3x-2} = \infty$$

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Past Exam Questions

Functions and Limits

Past-Exam Questions

Question 1 (1 pts) Compute the following limits:

a) **(0.5 pts)** $\lim_{x \to 1} \frac{\sqrt{x} - 1}{x - 1}$

b) **(0.5 pts)** $\lim_{x \to \infty} \frac{-3x^4 + 5x^2}{x^3 - x + 7}$

Question 2 (1.5 pts) Let $f(x) = \frac{\sin x}{x}$. Find $\lim_{x \to 0} f(x)$.

Hint: When x is close to 0, the function satisfies that $\cos x \leq f(x) \leq 1$.

Try and solve these on your own without looking at the answers. It is highly recommended that you participate in the exercises and the questions.

Past-Exam Questions

a) **(0.5 pts)** $\lim_{x \to 1} \frac{\sqrt{x} - 1}{x - 1}$

a) $\lim_{x \to 1} \frac{\sqrt{x} - 1}{x - 1}$

Try direct sub gives $\frac{0}{0}$, so we multiply by $\frac{\sqrt{x} + 1}{\sqrt{x} + 1}$

$$\lim_{x \to 1} \frac{\sqrt{x} - 1 \cdot \sqrt{x} + 1}{x - 1 \cdot \sqrt{x} + 1} = \lim_{x \to 1} \frac{(x - 1)}{(x-1)(\sqrt{x} + 1)} = \lim_{x \to 1} \frac{1}{\sqrt{x} + 1}$$

cancel of common factor

Sub in $x = 1$ $= \lim_{x \to 1} \frac{1}{1+1} = \frac{1}{2}$

Past-Exam Questions

b) **(0.5 pts)** $\lim_{x\to\infty} \dfrac{-3x^4+5x^2}{x^3-x+7}$

b) $\lim_{x\to\infty} \dfrac{-3x^4+5x^2}{x^3-x+7}$

Since this is a rational function whose numerator has a higher degree and thus will grow faster, the limit will be given by the leading term, putting into account the negative sign

$$\lim_{x\to\infty} \frac{-3x^4+5x^2}{x^3-x+7} = -\infty$$

* Remember trick 3

If you want to try to solve it, divide num and den by leading term x^4:

$$\frac{-3+5/x^2}{\frac{1}{x}-\frac{1}{x^3}+\frac{7}{x^4}} = \frac{-3}{0} = -\infty$$

$5/x^2 \to 0$

As $x\to\infty$, these approach 0

General rules / tricks:

If degree(top) < degree(bottom) → limit is **0**

If degree(top) = degree(bottom) → limit is **ratio of leading coefficients**

If degree(top) > degree(bottom) → limit is **±∞** (or does not exist)

Past-Exam Questions

Question 2 (1.5 pts) Let $f(x) = \dfrac{\sin x}{x}$. Find $\lim_{x \to 0} f(x)$.

Hint: When x is close to 0, the function satisfies that $\cos x \leq f(x) \leq 1$.

2) $f(x) = \dfrac{\sin x}{x}$

find $\lim_{x \to 0} f(x) \Rightarrow \lim_{x \to 0} \dfrac{\sin x}{x}$

We can use the squeeze theorem:

Since $\cos x \leq f(x) \leq 1$ when x is close to 0 and:

$\lim_{x \to 0} \cos x = 1$ and $\lim_{x \to 0} 1 = 1$

$\lim_{x \to 0} \dfrac{\sin x}{x} = 1$

* Use these theories to your advantage. Makes it much easier.

Theorem (1.2.4). ***The Squeeze Theorem*** *Let f, g, h functions such that $f(x) \leq g(x) \leq h(x)$ $\forall x \in (b, c)$ and let $a \in (b, c)$. Assume $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$. Then $\lim_{x \to a} g(x) = L$.*

Past-Exam Questions

4. **(1 pts.) Calculus:** Does $\lim_{x \to \frac{\pi}{2}} \tan x$ exist? Why?

Try and solve these on your own without looking at the answers. It is highly recommended that you participate in the exercises and the questions.

Past-Exam Questions

4. **(1 pts.) Calculus:** Does $\lim_{x \to \frac{\pi}{2}} \tan x$ exist? Why?

4) Does $\lim_{x \to \frac{\pi}{2}} \tan x$ exist?

To solve this, we need to see if the lateral limits coincide

$$\lim_{x \to \frac{\pi}{2}^-} \tan x = \lim_{x \to \frac{\pi}{2}^-} \frac{\sin x}{\cos x} = -\infty$$

Since ~~at~~ on the left of $\frac{\pi}{2}$, the cosine is negative and sin is 1 and approaches 0.

Now,

$$\lim_{x \to \frac{\pi}{2}^+} \tan x = \lim_{x \to \frac{\pi}{2}^+} \frac{\sin x}{\cos x} = +\infty$$

Since on the right, cosine is positive while approaching 0.

Since they do not coincide, the limit doesn't exist.

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Good Luck!